🌐 DataSphere — E-Commerce Sales & Customer Insights Platform

DataSphere is a backend data pipeline and analytics platform built on the real-world **Olist Brazilian E-Commerce dataset**. It ingests and integrates multiple data sources, stores and manages data using **AWS S3 and Snowflake**, and delivers business insights through a lightweight dashboard powered by automated pattern analysis.

🏗️ Architecture Overview

Raw CSV Data (Olist) → Data Ingestion & Preprocessing → AWS S3 (Storage) → Snowflake (Data Warehouse) → Analysis & Pattern Detection → Dashboard Visualization

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🚀 Features

- **Backend Data Pipeline** — Ingests, cleans, and integrates multiple Olist CSV datasets covering orders, customers, payments, reviews, sellers, products, and geolocation
- **AWS S3 Integration** — Connects to AWS S3 for cloud-based data storage and retrieval
- **Snowflake Data Warehouse** — Maintains a live, updatable database layer in Snowflake for scalable querying and storage
- **Automated Pattern Analysis** — Model-driven analysis identifies trends and patterns across sales, customer behavior, and seller performance
- **Insight Dashboard** — Visualizes key business metrics and analytical findings in a simple, readable format

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🛠️ Tech Stack

Layer	Technology
Language	Python
Data Processing	Pandas, NumPy
Cloud Storage	AWS S3
Data Warehouse	Snowflake
Analysis & Visualization	Matplotlib, Scikit-learn
Environment	Jupyter Notebook / VS Code

📂 Dataset

The project uses the **Olist Brazilian E-Commerce Public Dataset**, which includes the following files:

- `olist_orders_dataset.csv` — Order lifecycle and status
- `olist_customers_dataset.csv` — Customer identifiers and location
- `olist_order_items_dataset.csv` — Products and sellers per order
- `olist_order_payments_dataset.csv` — Payment methods and values
- `olist_order_reviews_dataset.csv` — Customer satisfaction scores and comments
- `olist_products_dataset.csv` — Product categories and attributes
- `olist_sellers_dataset.csv` — Seller identifiers and location
- `olist_geolocation_dataset.csv` — Geographic coordinates for customers and sellers
- `product_category_name_translation.csv` — Category name translations (Portuguese → English)

📊 Key Insights Delivered

- Sales trends across product categories and time periods
- Customer purchasing patterns and retention signals
- Seller performance and order fulfillment analysis
- Payment method distribution and order value breakdown
- Geographic distribution of customers and sellers across Brazil

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⚙️ Setup & Installation

1. Clone the repository

```bash
git clone https://github.com/Savani-Raj/DataSphere-Project.git
cd DataSphere-Project
```

2. Install dependencies

```bash
```

```
pip install -r requirements.txt
```

3. Configure AWS credentials

Set up your AWS credentials to enable S3 connectivity:

```bash
aws configure
```

4. Configure Snowflake connection

Update your Snowflake connection parameters (account, warehouse, database, schema) in the configuration file before running the pipeline.

5. Run the pipeline

```bash
python main.py
```

📁 Project Structure

```
DataSphere-Project/
|── data/                  # Raw Olist CSV datasets
|── src/                   # Pipeline and analysis source code
|── notebooks/             # Exploratory analysis notebooks
|── outputs/               # Generated visualizations and reports
|── requirements.txt       # Python dependencies
|── README.md              # Project documentation
```

📜 License

This project is licensed under the MIT License.

📧 Contact

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